

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY CHANGA

B.Tech. EE Sem - III Examination, 2011
EE – 202 Electrical Power Generation

Date: 30/11/2011, Wednesday

Time: 1.30 to 4.30 p.m

Maximum Marks: 70

Instructions:

1. Attempt all questions from both sections.
2. Use separate answer book for each sections.
3. Figures to the right indicate **full** marks.
4. Assume suitable additional data, if necessary

SECTION – I

- Q-1 Draw well labeled layout of steam power plant. 7
- Q-2 A. Explain coal handling plant for the coal fired thermal power plant. 6
- B. List out the equipments in proper order that comes in the water-steam cycle of steam power plant. Start from boiler drum. 2
- C. Why we condense the steam in condenser even if there are huge thermal loss in a condenser occurs? 3
- D. Why demineralised water is used for the production of steam inside the boiler drum? 3

OR

- Q-2 A. Explain construction and working of electro static precipitator for large thermal power plant. 6
- B. Write technical note on “wind farm siting”. 3
- C. List out the advantages of stoker firing over hand firing in connection with combustion equipment. 2
- D. List out the merits and de-merits of wind energy. 3
- Q-3 A. With well labelled diagram explain solar wind hybrid system. 3
- B. With well labelled diagram explain the working of open cycle MHD system. 5
- C. With well labelled diagrams explain the tidal energy conversion. 6

OR

- Q-3 A. State the location and function of following equipments in thermal power plants. 3
- (i) Ring Header
- (ii) Down comer
- (iii) Boiler Feed Pump
- B. Explain construction and working of alkaline H_2 - O_2 fuel cell. 5
- C. With well labeled diagram explain the working of open cycle OTEC plant. 6

SECTION – II

Q-4 Draw the schematic diagram of hydro-electric power station and explain the working of hydro-electric power station. 7

Q-5 A. State the advantages and dis-advantages of Hydro-electric power station. 5

B. Draw the proper diagram of nuclear reactor and explain briefly. 7

C. Explain about panstock in connection with Hydro-electric power plant. 2

OR

Q-5 A. Draw the schematic diagram of nuclear power station and explain the working of nuclear power station. 7

B. Draw the schematic diagram of gas turbine power plant and explain the working of gas turbine power plant. 7

Q-6 A. Which points should be taken into account while selecting the site for nuclear power station? 4

B. Explain simple flat plate collector for solar power plant. 5

C. Draw the neat diagram of solar pond and describe in details. 5

OR

Q-6 A. Explain classification and types of geo-thermal power plant. 4

B. Explain the difference between a geo-thermal power plant and a thermal power plant. 4

C. Draw the schematic diagram of Diesel power station and explain the working of Diesel-electric power station. 6